DNA Provides Evidence for Evolution



Comparative Genomics

Endogenous Retroviruses

Mitochondrial DNA

Overview

- Idea of evolution and evidence for it accepted by vast majority of scientists
- Has been controversial for reasons unrelated to the scientific evidence
- Important to look at evidence to support scientific theory
- Recent advances in technology provide further evidence to support the theory of evolution



Evidence for Evolution – Comparing DNA (comparative genomics)

- DNA
  - Uses sequences of 4 nitrogenous bases
  - Codes for protein production
  - All living things use the same 4 bases, the same DNA code – indication of common origin
  - Sequence of bases varies
    - Mutation
    - Natural selection
    - Genetic Drift
  - As living things reproduce, code differs over generations among offspring, despite common ancestor, but some stays the same.
    - More distantly related: greater variation
    - Closely related: more shared

Table 15.1 Relationship between humans and great apes using DNA differences

Primates being compared	DNA difference (%)
Human – chimpanzee	1.2
Chimpanzee – gorilla	1.2
Human – gorilla	1.6
Chimpanzee – orang-utan	1.8
Human – orang-utan	2.4
Gorilla – orang-utan	2.4

Learning Aim: Review DNA, protein production and variation.

Evidence for Evolution – Comparing DNA

- Chromosomes also contain sequences of bases that don't appear to code for anything:
  - Non-coding, or "junk DNA"
  - Also varies across generations due to random mutation.
  - Can also be *compared* for similarities and differences to assess relatedness.
- Other examples of DNA that can be compared to assess relatedness
  - Endogenous retroviruses (ERV)
  - Mitochondrial DNA

Learning Aim: List types of DNA that can be compared to assess relatedness between organisms.

Comparing DNA - Endogenous Retroviruses

- Retroviruses are viruses that incorporate their DNA code into the cells of the infected organism. Some don't cause illness.
- Endogenous retroviruses are viral sequences that have become part of the genome in the germline cells of the host organism.
- They are therefore passed on to each new generation
- Make up 8% of human genome
- Other primates have some of the same sequences in the same locations, indicating that we, and they, are descended from a common ancestor who was originally infected with the virus.
- Can map this (see diagram) to show relationships in terms of common ancestors, between modern species.

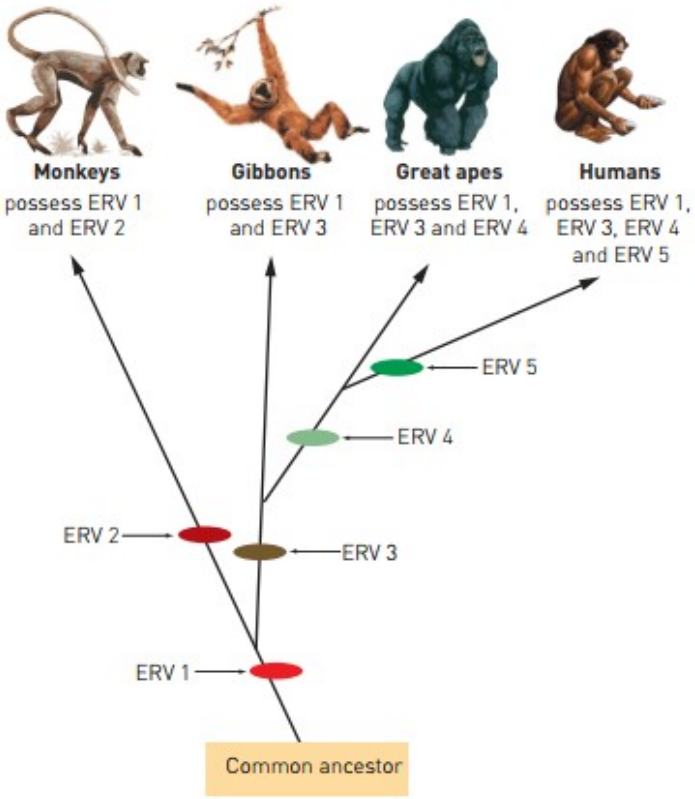


Figure 15.1 A simplified example of how endogenous retroviruses could be used to trace common ancestry: the great apes and humans have a more recent common ancestor as they share more endogenous retroviruses.

Note: In order to compare this, we need to sequence the genome using the techniques previously discussed:

- PCR to amplify sample
- Sequencing
- Electrophoresis

Learning Aim: Describe how endogenous retroviruses can be used to map relatedness between organisms.

Comparing DNA- mtDNA

- mtDNA = mitochondrial DNA. From mitochondria which has it's own DNA separate to the nuclear DNA
- Small circular molecules
  - mtDNA: 37 genes, all essential,
  - 24 code for tRNA
- Inherited from mother – in cytoplasm of ovum
  - Can therefore trace female ancestral line
  - mtDNA has slowly diverged via random mutation from common female ancestor "Mitochondrial Eve" (note the mutations do not cause impairment of function – they are point mutations that do not change the amino acids produced)
- Can compare level of similarity between individuals to see how long ago their last common ancestor lived.
- Also used to track migration routes of ancient humans

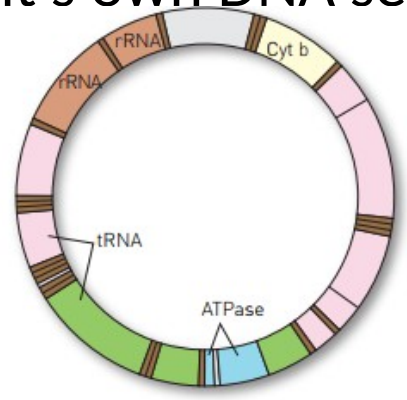
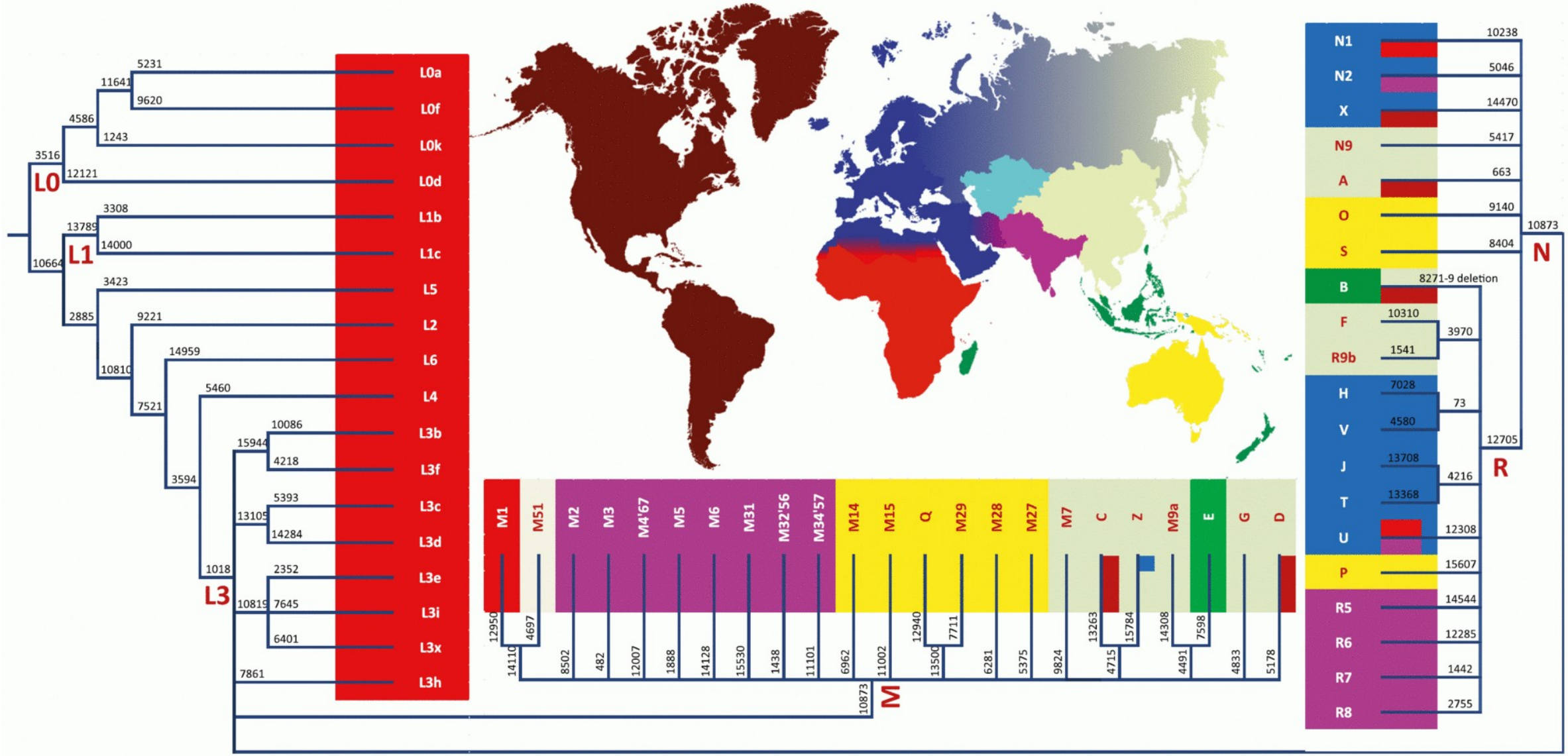
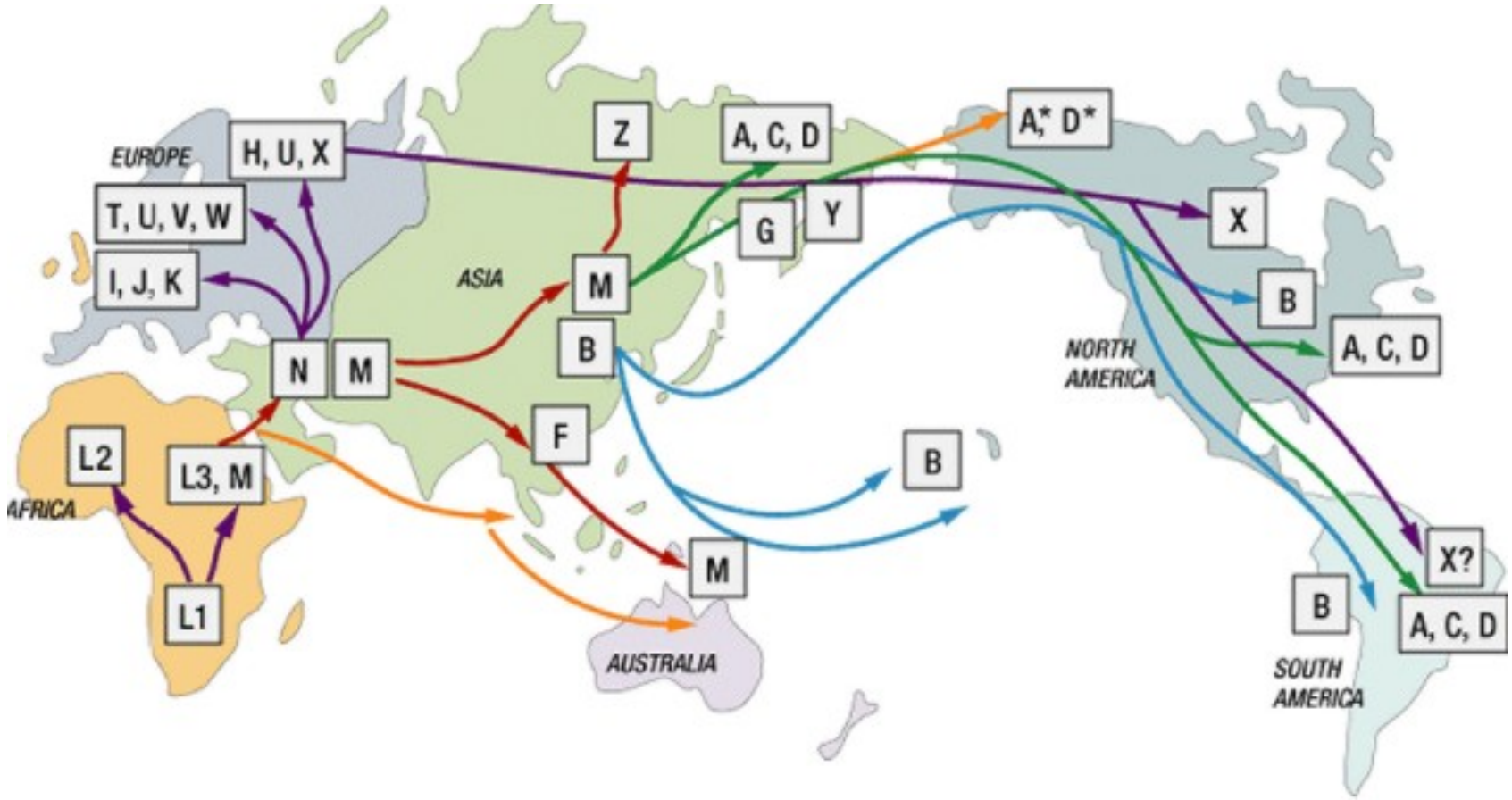


Figure 15.2 A model of a molecule of mtDNA showing the location of some of the genes

Learning Aim: Describe how mtDNA can be used to map relatedness between organisms.



mtDNA haplogroup tree and distribution map.[3] The numbers are haplogroup labels, reported according to the http://www.phylotree.org/ nomenclature,[4] and give the location of one of the mutations leading to the derived haplotype. (Only a single branch defining marker, preferably from the coding region, is shown.) The main geographic features of haplogroup distribution are highlighted with colour.



EXPANSION TIMES *(years ago)*	
Africa	120,000 - 150,000
Out of Africa	55,000 - 75,000
Asia	40,000 - 70,000
Australia/PNG	40,000 - 60,000
Europe	35,000 - 50,000
Americas	15,000 - 35,000
Na-Dene/Esk/Aleuts	8,000 - 10,000

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mtDNA Migrations Map

- mtDNA: important tool in mapping relationships between species
  - Verifies other evidence
  - eg: examination shows that last common ancestor of modern humans and neanderthals lived around 600 000 years ago.
  - (This does not mean that the last neanderthal was then. Why?)

Learning Aim: Describe how mtDNA can be used to map relatedness between organisms.